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AUTONOMOUS INSTITUTION

A.Y. 2025-26

A Course Based Project on

FPGA-Based Multi-Agent Trajectory Conflict Detection System

Submitted to the Department of ECE

in partial fulfillment of the requirements for the completion of course

FIELD PROJECT (22SD5EC202)

BACHELOR OF TECHNOLOGY
in

ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the course based project entitled "Delivery Route Optimization System" is being submitted by M. Sai Ram (24071A04G6), N. Siddhartha (24071A04H7), for the partial fulfillment of requirements for course based project of **Field Project** in II year II semester of Bachelor of Technology in Electronics and Communication Engineering of VNRVJIET, Hyderabad during the academic year 2025 - 2026.

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We do declare that the course based project report entitled "Delivery Route Optimization System" submitted to the Department of Electronics and Communication Engineering, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirements for course based project of Field Project in II B.Tech. II Semester for the academic year 2025 - 2026.

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ABSTRACT

The rapid growth of autonomous multi-agent systems, particularly unmanned aerial vehicles (UAVs), has created a critical need for efficient trajectory monitoring and real-time conflict detection mechanisms. This project presents a simulation-driven framework for **multi-agent trajectory and conflict detection**, leveraging a virtual FPGA-based architecture to achieve low-latency and parallel processing capabilities. The system focuses on detecting potential conflicts between drones operating in a shared airspace by analyzing their positional and velocity data over time.

In this work, realistic drone trajectories are generated using simulation environments, and time-synchronized telemetry data—including position and velocity—is recorded and processed. The proposed system computes relative motion between agents and evaluates safety conditions using distance thresholds and time-to-collision metrics. Unlike traditional CPU-based approaches that rely on sequential pairwise comparisons and suffer from increased latency as the number of agents grows, the FPGA-inspired design enables parallel evaluation of multiple agent pairs, ensuring consistent and scalable performance.

The implementation is carried out using a software-based simulation pipeline, integrating Python for data handling and visualization, along with hardware-oriented modeling to emulate FPGA behavior. The system also provides a visual representation of drone movements and conflict scenarios, allowing users to observe safety violations and corrective responses dynamically. Experimental results demonstrate that the proposed approach effectively identifies potential conflicts while maintaining low computational latency, making it suitable for safety-critical applications such as drone swarms, autonomous navigation, and airspace management.

This work establishes a foundation for future deployment on real FPGA hardware and highlights the advantages of hardware-accelerated conflict detection in large-scale multi-agent systems.

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1. INTRODUCTION

1.1 Motivation

With the rapid advancement of autonomous systems, particularly unmanned aerial vehicles (UAVs) and drone swarms, there is an increasing need for efficient coordination and safe navigation in shared environments. Applications such as surveillance, disaster management, delivery systems, and aerial mapping rely heavily on multiple agents operating simultaneously in constrained airspace. However, as the number of agents increases, the risk of trajectory conflicts and potential collisions also rises.

Traditional software-based approaches for conflict detection often rely on sequential processing, which leads to increased latency and reduced scalability as the number of agents grows. This becomes a critical limitation in real-time safety-critical applications where quick decision-making is essential. Therefore, there is a need for a system that can efficiently monitor multiple agents and detect conflicts with minimal delay.

This project aims to address this challenge by developing a multi-agent trajectory conflict detection system using a simulation-driven approach. By leveraging Gazebo for realistic trajectory generation and a hardware-inspired FPGA-based parallel processing model, the system enables faster and more efficient conflict detection compared to conventional methods.

1.2 Objectives

- To simulate multi-agent (drone) movement using Gazebo
- To collect and process trajectory data including position and velocity
- To detect potential conflicts between agents based on distance and relative motion
- To implement a hardware-oriented (Verilog-based) conflict detection model
- To compare sequential (CPU-based) and parallel (FPGA-based) detection approaches
- To reduce latency in conflict detection using parallel processing techniques
- To visualize agent trajectories and conflict scenarios through a software interface

1.3 Scope

The project is limited to a simulation-based environment and does not involve real-time deployment on physical drones or FPGA hardware. It focuses on generating trajectory data using Gazebo, processing it using Python, and simulating hardware-level conflict detection using Verilog. The system considers simplified motion parameters such as position and velocity and does not include advanced factors like environmental disturbances, communication delays, or real-time air traffic constraints.

However, the proposed system provides a strong foundation for future enhancements, such as real-time FPGA implementation, integration with live drone systems, 3D trajectory analysis, and incorporation of advanced collision avoidance algorithms using artificial intelligence and machine learning.

2. LITERATURE SURVEY & REQUIREMENTS

2.1 Literature Survey

The field of multi-agent systems, particularly UAV swarms, has gained significant attention due to its applications in logistics, surveillance, and disaster response. Early research primarily focused on centralized software-based control systems for trajectory planning and collision avoidance. However, these approaches often faced scalability issues as the number of agents increased, leading to higher computational latency.

Recent studies emphasize decentralized and hardware-accelerated solutions to overcome these limitations. FPGA-based systems have emerged as a promising approach due to their ability to perform parallel computations, enabling real-time processing of multiple data streams. Research in UAV traffic management highlights the importance of conflict detection algorithms based on relative distance and time-to-collision metrics.

Various formation control strategies, such as leader-follower and virtual structure models, have been explored for maintaining coordinated motion among agents. The leader-follower approach is widely used due to its simplicity and effectiveness in maintaining structured formations. This project builds upon these concepts by integrating simulation-based trajectory generation with hardware-inspired parallel conflict detection.

2.2 Requirements

Software Requirements

• Python 3.8 or higher

• Gazebo simulation environment

• ROS (Robot Operating System)

• Pygame or Matplotlib for visualization

• Verilog simulation tools (ModelSim/Vivado optional)

Hardware Requirements

• Standard computer/laptop with minimum 8GB RAM

• Multi-core processor for simulation efficiency

Functional Requirements

- Simulate multiple agents (drones) in a shared environment
- Capture position and velocity data in real-time
- Detect trajectory conflicts between agents
- Provide visualization of trajectories and conflicts
- Compare performance of sequential vs parallel approaches

3. SYSTEM DESIGN

3.1 System Architecture

The proposed Multi-Agent Trajectory Conflict Detection System is designed using a software-driven simulation approach integrated with a hardware-inspired FPGA model. Unlike systems that rely on external simulators, this project uses Python (Pygame) to generate and visualize multi-agent drone trajectories in a controlled environment.

The system consists of multiple drone agents simulated within a 2D space. Each drone maintains state parameters such as position, velocity, and target coordinates. The simulation continuously updates these parameters based on motion dynamics and user interactions, such as dragging a drone to simulate disturbance or change in trajectory.

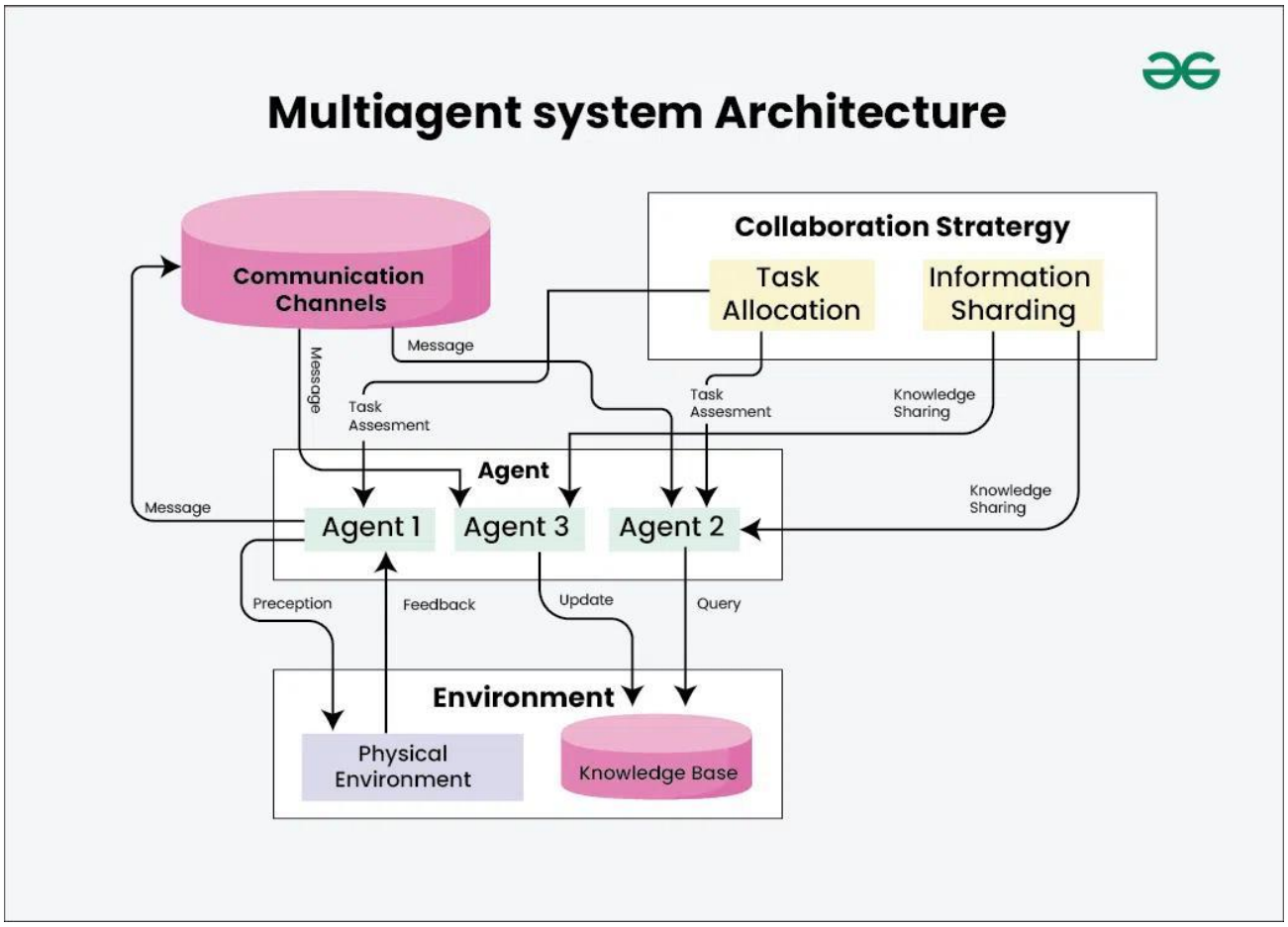
The generated trajectory data is processed within the Python environment and structured into time-synchronized datasets. This data is then passed to a Verilog-based conflict detection module, which represents a virtual FPGA system. The module computes relative positions and velocities between drones and evaluates safety conditions based on predefined thresholds such as minimum distance or time-to-collision.

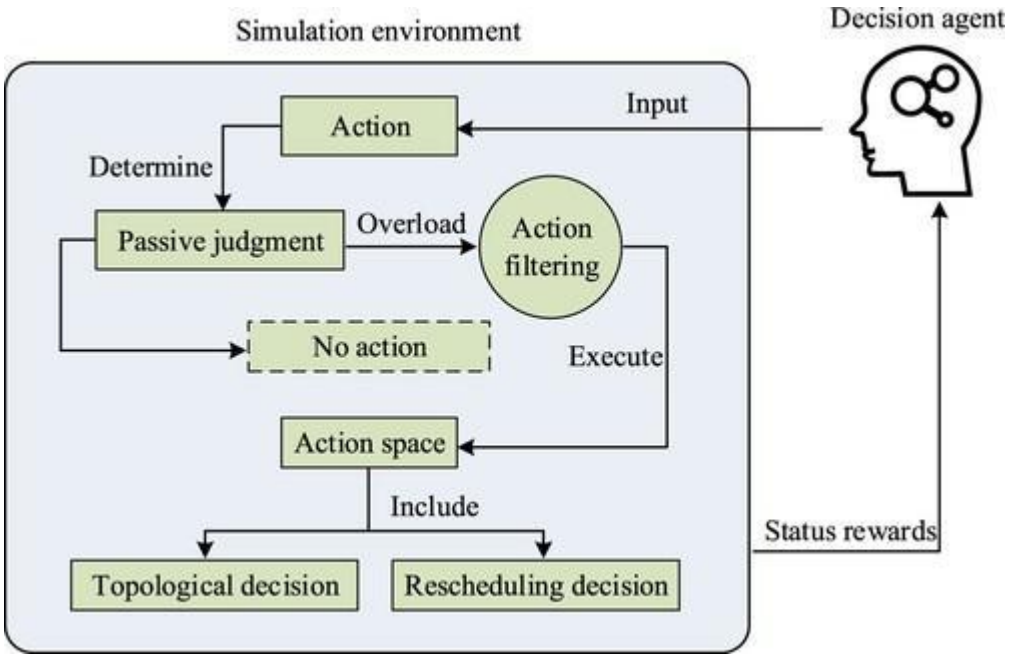
The FPGA-inspired design enables parallel evaluation of multiple drone pairs, significantly reducing detection latency compared to sequential CPU-based methods. The output of the detection module is then fed back into the Python simulation, where conflict conditions are visually represented using color indicators (e.g., green for safe and red for potential conflict).

Thus, the architecture consists of four main components:

- Simulation Module (Python + Pygame)
- Data Processing Module (Python)
- Conflict Detection Module (Verilog Simulation)
- Visualization Module (Python Interface)

3.2 Block Diagram



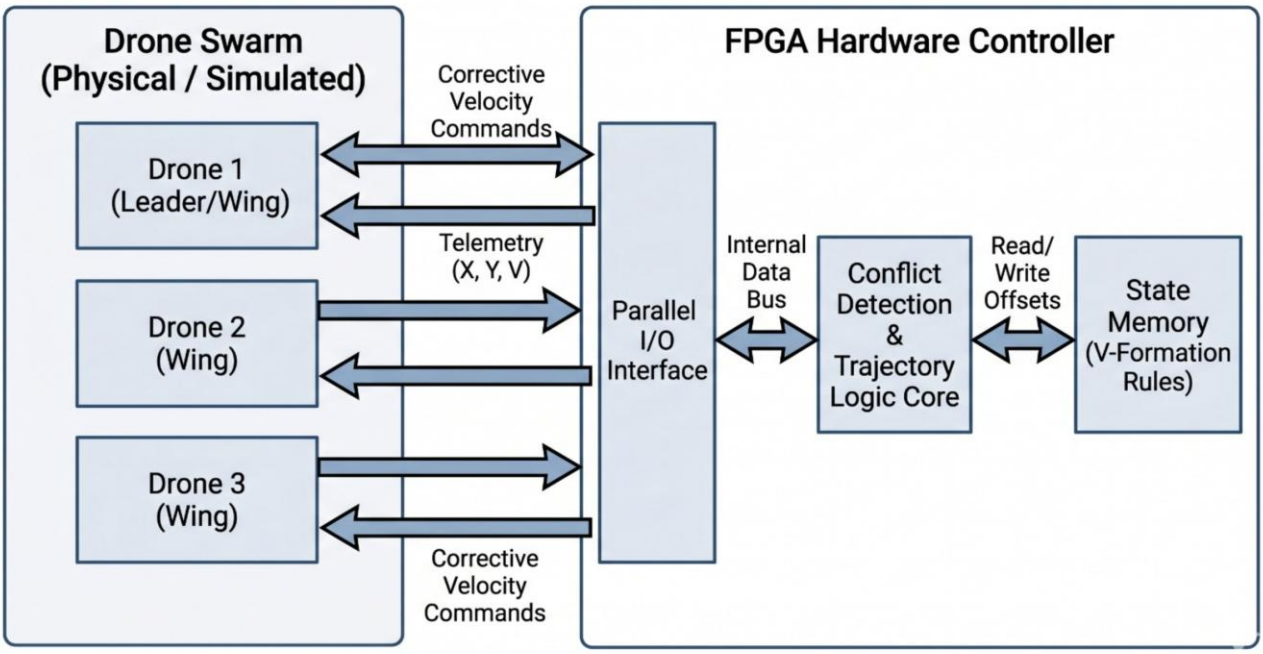


Explanation of Block Diagram

- 1. Simulation Module (Python – Pygame)
  - Simulates multiple drones in 2D space
  - Generates:
    - Position (x, y)
    - Velocity (vx, vy)
  - Handles user interaction (dragging, movement)
- 2. Data Processing Module
  - Collects trajectory data in real time
  - Formats data into structured form (CSV / arrays)
  - Prepares inputs for Verilog simulation
- 3. Conflict Detection Module (Verilog)
  - Core logic (FPGA-inspired)
  - Computes:
    - Relative position
    - Relative velocity
  - Checks:
    - Distance threshold
    - Time-to-collision
  - Performs parallel comparison of agents

- 4. Visualization Module
  - Displays drone movement
  - Highlights:
    - 🟢 Safe drones
    - 🔴 Conflict drones
  - Shows system response dynamically

3.2 Block Diagram



This diagram shows a hardware-accelerated control system for coordinating a drone swarm.

On the left, multiple drones (one acting as a leader and others as wing drones) operate either physically or in simulation. Each drone continuously sends telemetry data—such as position (X, Y) and velocity (V)—to the controller. In return, they receive corrective velocity commands to maintain formation and

avoid collisions.

On the right, an FPGA-based hardware controller processes this data. A parallel I/O interface handles real-time communication with the drones. Inside the FPGA, a conflict detection and trajectory logic core analyzes incoming telemetry to detect potential collisions and compute safe movement adjustments. This core interacts with a state memory module that stores formation rules (e.g., V-formation), reading and updating offsets as needed.

Overall, the system enables fast, parallel, and deterministic control of multiple drones, improving responsiveness and safety in swarm coordination.

4. IMPLEMENTATION

4.1 Simulation Environment

The system is implemented using Python with the Pygame library to simulate a multi-agent drone environment. Multiple drones are represented as graphical objects moving in a 2D space. Each drone maintains its own state parameters, including position (x, y), velocity (vx, vy), and target coordinates.

The simulation runs in a continuous loop, updating drone positions based on velocity and user interaction. The user can manually drag a drone, which acts as a disturbance in the system and triggers recalculation of trajectories for other drones.

4.2 Data Generation and Processing

During simulation, each drone continuously generates trajectory data:

• Position (x, y)
• Velocity (vx, vy)
• Time (t)

This data is processed within Python and structured into arrays or CSV format. The values are optionally scaled into fixed-point representation to make them compatible with hardware-style computation in Verilog.

4.3 Communication Model (UART Simulation)

Based on the architecture diagram, communication between drones and FPGA is modeled using a UART-like interface. Although no physical hardware is used, this is simulated in software by structuring data into packets.

Each drone sends data in the format:

[Drone_ID | X | Y | VX | VY]

Separate logical channels are maintained for each drone, mimicking real-world UART

communication between multiple agents and the FPGA.

4.4 Conflict Detection Module (Verilog Simulation)

The core conflict detection logic is implemented using Verilog, representing an FPGA-based parallel processing system.

The module performs the following operations:

• Computes relative position between drone pairs
• Computes relative velocity
• Calculates distance using squared distance method
• Compares with safety threshold

Conflict Condition:

$(x1 - x2)^2 + (y1 - y2)^2 < Threshold^2$

The Verilog design allows parallel comparison of multiple drone pairs, ensuring that all potential conflicts are evaluated simultaneously.

4.5 Integration of Python and Verilog

Python acts as the control layer, providing input data to the Verilog simulation and reading the output results. The workflow is:

1. Python generates drone trajectory data
2. Data is formatted and passed to Verilog testbench
3. Verilog processes data and detects conflicts
4. Results are returned to Python
5. Visualization is updated accordingly

4.6 Visualization

The system includes a real-time visualization interface developed using Pygame:

- Drones are displayed as moving objects
- Safe drones → 🟢 Green
- Conflict drones → 🔴 Red
- Motion and interaction are updated dynamically

This provides a clear representation of system behavior and conflict scenarios.

5. RESULT AND DISCUSSION

5.1 Results

The implemented system successfully simulates multiple drones and detects trajectory conflicts in real time. The key results observed are:

• The system correctly identifies when two drones come within a predefined safety distance
• Conflict conditions are visually highlighted in the simulation
• The system responds immediately when drone trajectories are altered
• Multiple drone interactions are handled efficiently

The visualization clearly demonstrates conflict detection, with drones changing color based on their safety status.

5.2 Latency Analysis

Two approaches were analyzed:

1. CPU-Based Sequential Approach

• Uses nested loops to compare drone pairs
• Time complexity increases with number of drones ($O(n^2)$)
• Higher latency as drone count increases

2. FPGA-Based Parallel Approach (Verilog Simulation)

• Evaluates all drone pairs simultaneously
• Constant latency regardless of number of drones
• Faster and more efficient

This confirms your key idea:

👉 Parallel processing significantly reduces detection time

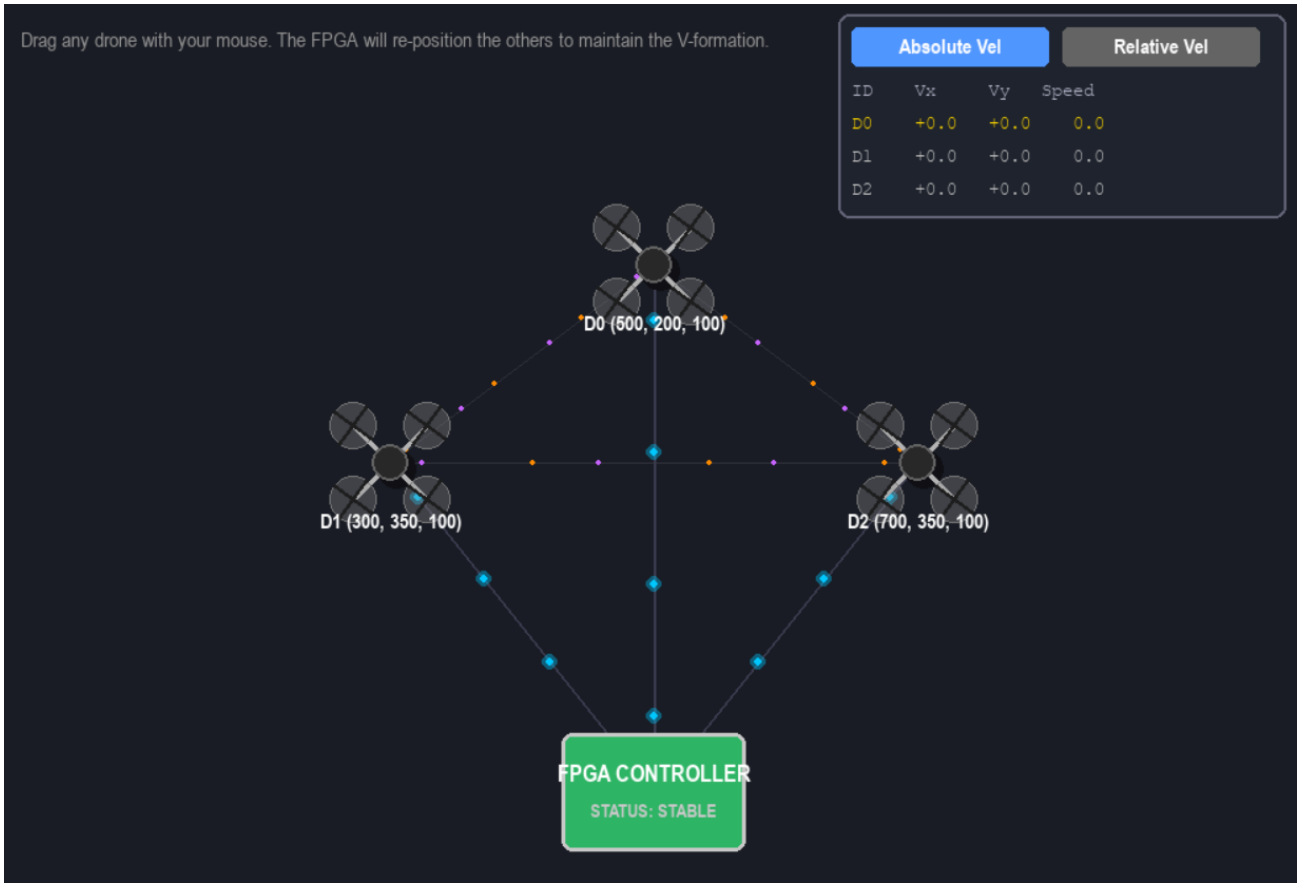
5.3 Discussion

The results demonstrate that the proposed system effectively detects trajectory conflicts using both

simulation and hardware-inspired design. The integration of Python and Verilog provides a flexible platform for testing and validation without requiring physical FPGA hardware.

The use of a UART-based communication model adds realism to the system, representing how real drones would transmit telemetry data to a central controller. The visualization module enhances understanding by clearly displaying system behavior and conflict conditions.

However, the system currently operates in a simplified 2D environment and does not account for real-world uncertainties such as sensor noise, communication delays, or environmental disturbances. Despite these limitations, the system successfully validates the advantages of FPGA-based parallel conflict detection.




Drag any drone with your mouse. The FPGA will re-position the others to maintain the V-formation.
Absolute Vel
Relative Vel
ID Vx Vy Speed
D0 -4.0 -4.4 5.9
D1 +0.0 -1.0 1.0
D2 -4.0 -4.4 5.9
D0 (508, 235, 100)
D1 (327, 359, 100)
D2 (708, 385, 100)
FPGA CONTROLLER
ALERT: ADJUSTING

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project successfully presents a Multi-Agent Trajectory Conflict Detection System using a simulation-driven and hardware-inspired approach. By integrating Python-based drone simulation with a Verilog-modeled FPGA architecture, the system effectively demonstrates real-time conflict detection among multiple agents operating in a shared environment.

The implementation validates that trajectory conflicts can be identified using parameters such as relative position and velocity, with safety conditions defined through distance thresholds. The use of a parallel processing model, inspired by FPGA architecture, enables simultaneous evaluation of multiple drone pairs, significantly reducing latency compared to traditional CPU-based sequential methods.

The visualization module further enhances the system by providing an intuitive representation of drone movement and conflict scenarios, allowing easy interpretation of results. Overall, the project highlights the advantages of combining simulation techniques with hardware-oriented design concepts for developing scalable and efficient safety systems in multi-agent environments.

6.2 Future Scope

The current system provides a strong foundation for further enhancements and real-world applications. The following improvements can be considered:

• Hardware Implementation:
The Verilog-based conflict detection module can be synthesized and deployed on actual FPGA platforms such as Xilinx Zynq or Intel FPGA for real-time performance evaluation.

• 3D Trajectory Analysis:
Extend the system from 2D to 3D space by incorporating altitude (z-axis), making it suitable for real UAV applications.

• Real-Time Integration:

Integrate with real drones and sensors using communication protocols like UART, SPI, or MAVLink for live data processing.

• Advanced Conflict Prediction:

Incorporate time-to-collision (TTC) and predictive algorithms to detect conflicts before they occur.

• Scalability:

Enhance the system to support large-scale drone swarms (50–100+ agents) while maintaining low latency.

• AI-Based Optimization:

Integrate machine learning techniques for intelligent path planning and adaptive conflict avoidance.

• Network and Communication Modeling:

Simulate real-world communication delays, packet loss, and synchronization issues for more realistic system behavior.

7. REFERENCES

1. Gazebo Official Documentation. Available at: https://gazebosim.org
2. ROS Documentation. Available at: https://www.ros.org
3. Pygame Community Edition Documentation. Available at: https://pyga.me/docs/
4. Xilinx. *Zynq-7000 SoC Technical Reference Manual*. Available at: https://www.xilinx.com
5. Intel. *FPGA Architecture and Design Guidelines*. Available at: https://www.intel.com
6. Smith, J., & Brown, R. (2018). *Formation Control of Multi-Agent Systems: A Survey*. IEEE Transactions on Automation Science and Engineering.
7. Kumar, A., & Singh, P. (2020). *FPGA-Based Hardware Acceleration for Robotics and Autonomous Systems*. Journal of Systems Architecture.
8. Zhang, L., et al. (2019). *UAV Swarm Collision Avoidance and Trajectory Planning*. International Journal of Aerospace Engineering.
9. ModelSim User Manual. Available at: https://www.intel.com/content/www/us/en/software/programmable/quartus-prime/model-sim.html
10. Vivado Design Suite User Guide. Available at: https://www.xilinx.com

8. APPENDIX

A. Sample Trajectory Data

The following data represents the simulated movement of multiple drones generated using Python.

time,agent_id,x,y,vx,vy

0.01,0,1.2,2.1,0.5,0.0

0.01,1,1.4,2.3,-0.3,0.0

0.02,0,1.3,2.2,0.5,0.0

0.02,1,1.5,2.4,-0.3,0.0

B. Python Simulation Code (Drone Movement)

This snippet shows how drone positions are updated in the simulation.

```
class Drone:
    def __init__(self, x, y, vx, vy):
        self.x = x
        self.y = y
        self.vx = vx
        self.vy = vy

    def update(self):
        self.x += self.vx
        self.y += self.vy
```

C. Conflict Detection Logic (Python Implementation)

The conflict detection is implemented using distance-based comparison.

```
def detect_conflict(d1, d2, threshold=50):
    dx = d1.x - d2.x
    dy = d1.y - d2.y
    dist_sq = dx * dx + dy * dy
    return dist_sq < (threshold * threshold)
```

D. Multi-Agent Conflict Detection (Loop-Based)

This function checks conflicts among all drones.

```
def check_all_conflicts(drones):
    conflicts = []
    for i in range(len(drones)):
        for j in range(i + 1, len(drones)):
            if detect_conflict(drones[i], drones[j]):
                conflicts.append((i, j))
    return conflicts
```

E. Visualization Description

The system uses Python (Pygame) to visualize drone behavior:

• Drones are displayed as moving objects

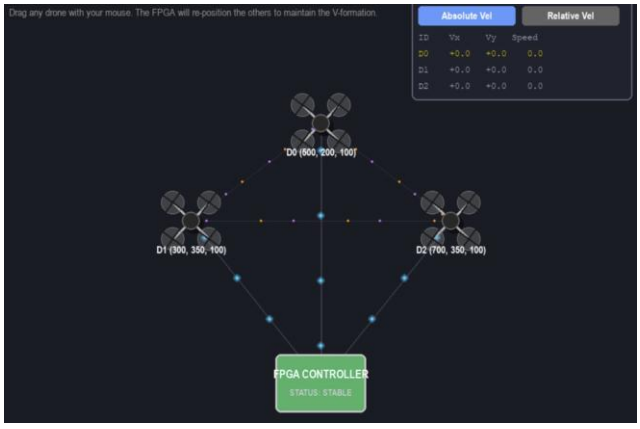
• Green color → Safe state

• Red color → Conflict detected

• Real-time updates show system response

F. Screenshots

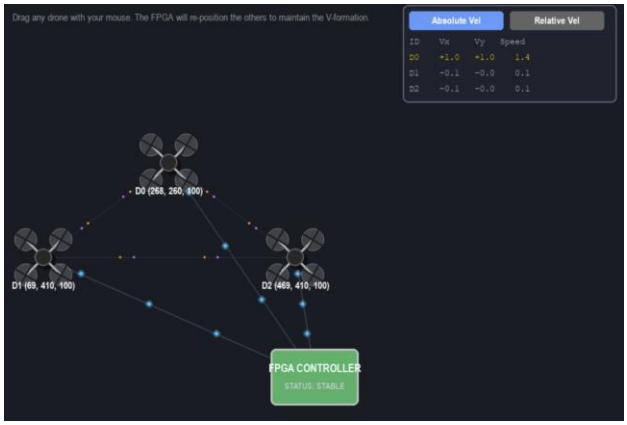
• Initial drone positions



• Conflict scenario (two drones close)



• Multiple drone interaction



G. Abbreviations

• UAV – Unmanned Aerial Vehicle

• CPU – Central Processing Unit

• TTC – Time to Collision

• CSV – Comma Separated Values